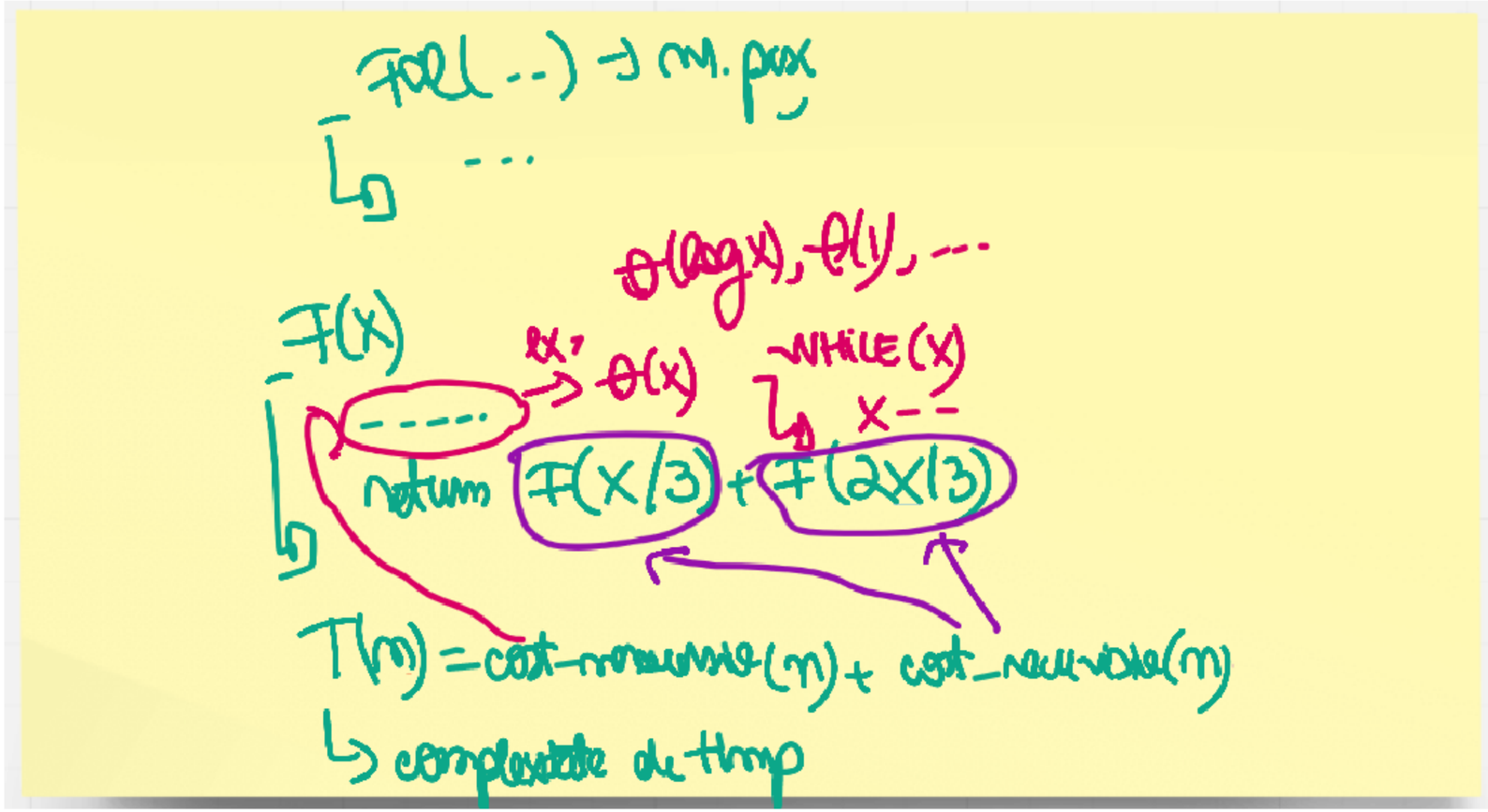
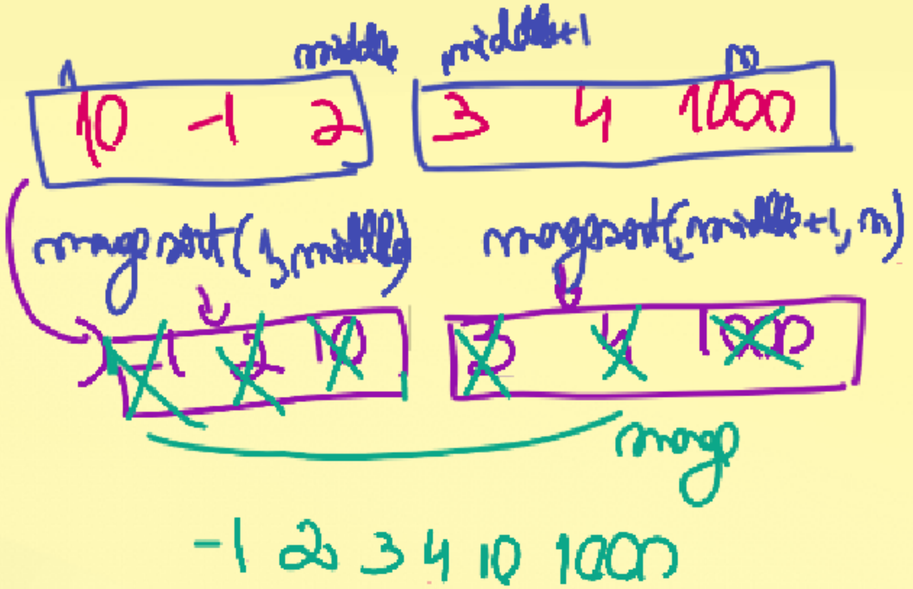

Fct(...) -> m.pour
...
θ(log x), θ(1), ...
F(x)

θ(x)
WHILE (x)
x--
return F(x/3) + F(2x/3)
T(n) = cost_non_recursive(n) + cost_recursive(n)
↳ complexité du temp



MERGE(v_1, v_2) → n elemente
↓
m elemente

$\Theta(m+n)$ = cost-recursiv

$c_1 g(n) \leq f(n) \leq c_2 g(n)$

$\Theta(n)$ = cost-recursiv

MERGESORT $(1, m)$
MERGESORT $(m+1, n)$
} cost-recursiv $\frac{n}{2}$

$$T(n) = T\left(\frac{n}{2}\right) + T\left(\frac{n}{2}\right) + \Theta(n) \rightarrow \text{e.m, m}$$

$$\boxed{T(n) = 2T\left(\frac{n}{2}\right) + n}$$

1) METODA ITERAȚIEI

$T(n) = 2T\left(\frac{n}{2}\right) + n$

$2T\left(\frac{n}{2}\right) = 2 \cdot 2T\left(\frac{n}{2^2}\right) + 2 \cdot \frac{n}{2}$

$2^2 T\left(\frac{n}{2^2}\right) = 2^3 T\left(\frac{n}{2^3}\right) + 3 \cdot \frac{n}{2^3}$

$2^k T\left(\frac{n}{2^k}\right) = 2^{k+1} T\left(\frac{n}{2^{k+1}}\right) + k \cdot \frac{n}{2^k}$

$2^h T\left(\frac{n}{2^h}\right) = 2^{h+1} T(1) + n$

$\frac{n}{2^{h+1}} = 1 \Rightarrow n = 2^{h+1} \Rightarrow h = \log n - 1$

$T(n) = 2^{h+1} T(1) + n(h+1) = \underbrace{2^{\log n}}_{n} + n \log n \in \Theta(n \log n)$

$T(1) = 1$

$\in \Theta(1)$

$T(n) = 2T\left(\frac{n}{2}\right) + n$

1) regola generale $T(f(n))$

ripeto $n = \frac{n}{2} \Rightarrow T\left(\frac{n}{2}\right) = 2T\left(\frac{n}{2^2}\right) + \frac{n}{2}$

$n = \frac{n}{2^2} \Rightarrow T\left(\frac{n}{2^2}\right) = 2T\left(\frac{n}{2^3}\right) + \frac{n}{4}$

$n = \frac{n}{2^k} \Rightarrow T\left(\frac{n}{2^k}\right) = 2T\left(\frac{n}{2^k \cdot 2}\right) + \frac{n}{2^k}$

2) condizione di uscita

$T(1) = 1 \in \Theta(1)$

$\frac{n}{2^{k+1}} = 1 \Rightarrow n = 2^{k+1}$

$\log n = k+1$

$\Rightarrow k = \log n - 1$

3)

$T(n) = 2T\left(\frac{n}{2}\right) + n$ ---- cuadro 0

$2T\left(\frac{n}{2}\right) = 2^2 T\left(\frac{n}{2^2}\right) + \frac{n}{2} \cdot 2$ — " — 1

$2^2 T\left(\frac{n}{2^2}\right) = 2^3 T\left(\frac{n}{2^3}\right) + 2^2 \frac{n}{2^2}$ — " — 2

$2^k T\left(\frac{n}{2^k}\right) = 2^{k+1} T\left(\frac{n}{2^{k+1}}\right) + 2^k \frac{n}{2^k}$ — " — k

$2^h T\left(\frac{n}{2^h}\right) = 2^{h+1} T\left(\frac{n}{2^{h+1}}\right) + 2^h \frac{n}{2^h}$ — " — h

$h+1$

$\frac{n}{2^{h+1}} = 1$

$T(n) = \underbrace{n + \dots + n}_{h+1 \text{ volte}} + 2^{h+1}\underbrace{T(1)}_{1} = 2^{h+1} + n(h+1)$

$h = \log n - 1 \Rightarrow T(n) = \underbrace{2^{\log n}}_{n} + n\log n = n_0 + n\log n \in \Theta(n\log n)$

$n \in O(n\log n)$

$a^{\log_b c} = c^{\log_b a}$

$2^{\log n} = 2^{\log_2 n} = n^{\log_2 2} = n^1$

$= n$

3) $T(n) = T(n-k) + \theta(n)$, $n \in \theta(n)$

$T(n) = T(n-k) + n$ (n − 0·k)

$T(n-k) = T(n-2k) + n-k$

$T(n-2k) = T(n-3k) + n-2k$

$T(n-ik) = T(n-(i+1)k) + n-ik$

$T(n-hk) = T(n-(h+1)k) + n-hk$

$T(1) = 1 \iff n-(h+1)k = 1 \iff h+1 = \frac{n-1}{k} \Rightarrow h = \frac{n-1}{k} - 1 \in \theta\left(\frac{n}{k}\right)$

$T(n) = T(1) + n + (n-k) + (n-2k) + \dots + (n-hk) =$

$= \sum_{i=0}^{h} (n-ik) = \sum_{i=0}^{h} n - \sum_{i=0}^{h} ik = n\sum_{i=0}^{h} 1 - k\sum_{i=0}^{h} i =$

$0+k+2k+\dots = k(0+1+2+\dots)$

$\sum_{i=0}^{h} 1 \to h+1$; $\sum_{i=0}^{h} i \to \frac{h(h+1)}{2}$

$\sum (a_n + b_n) = \sum a_n + \sum b_n$

$= n(h+1) - k \cdot \frac{h(h+1)}{2} = n \cdot \frac{n-1}{k} - k \cdot \frac{1}{2}\left(\frac{n-1}{k} - 1\right) \cdot \frac{n-1}{k} =$

$h = \frac{n-1}{k} - 1 \Rightarrow h+1 = \frac{n-1}{k}$

$= \frac{n(n-1)}{k} - \frac{1}{2}(n-1) \cdot \frac{n-1-k}{k} \in \theta\left(\frac{n^2}{k}\right) \in \theta(n^2)$

$n^2 - n \in \theta(n^2)$; k constantă

*) $T(n) = 2T(\sqrt{n}) + 1$

$T(\sqrt{n}) = 2T(\sqrt[4]{n}) + 1$

$T(n^{\frac{1}{2}}) = 2T(n^{\frac{1}{4}}) + 1$

$T(n^{\frac{1}{4}}) = 2T(n^{\frac{1}{8}}) + 1$

$T(n^{\frac{1}{8}}) = 2T(n^{\frac{1}{16}}) + 1$

$n, n^{\frac{1}{2}}, n^{\frac{1}{4}}, \ldots, n^{\frac{1}{2^{k-1}}}, \ldots, 1$

$n^{\frac{1}{2^{k+1}}} = 1 \Rightarrow \log n^{\frac{1}{2^{k+1}}} = \log 1$

$\frac{1}{2^{k+1}} \log n = 0$

$k \to \infty$

$T(2) = 1 \Rightarrow n^{\frac{1}{2^{k+1}}} = 2 \Rightarrow \frac{1}{2^{k+1}} \log n = \log 2$

$\frac{1}{2^{k+1}} = \frac{\log 2}{\log n} \Rightarrow 2^{k+1} = \frac{\log n}{\log 2}$ / log

$k+1 = \log\left(\frac{\log n}{\log 2}\right)$

$T(c) = 1$

$c > 1$

$T(n) = 2T(n^{\frac{1}{2}}) + 1$

$2T(n^{\frac{1}{2}}) = 2^2 T(n^{\frac{1}{2^2}}) + 2$

$2^2 T(n^{\frac{1}{2^2}}) = 2^3 T(n^{\frac{1}{2^3}}) + 2^2$

$2^k T(n^{\frac{1}{2^k}}) = 2^{k+1} T(n^{\frac{1}{2^{k+1}}}) + 2^k$

$2^{k+1} T(n^{\frac{1}{2^{k+1}}}) = 2^{k+1} T(n^{\frac{1}{2^{k+1}}}) + 2^k$

$c = 1$

$T(n) = 2^{k+1} + 1 + 2 + \ldots + 2^k + \ldots + 2^0$

$= \sum_{i=0}^{k+1} 2^i = \frac{2^{k+2} - 1}{2 - 1} = 2^{k+2} - 1 = \left(\frac{2}{\log 2}\right) \cdot \log n - 1 \in \Theta(\log n)$

$\sum_{i=0}^{m} a^i = \frac{a^{m+1} - 1}{a - 1}$

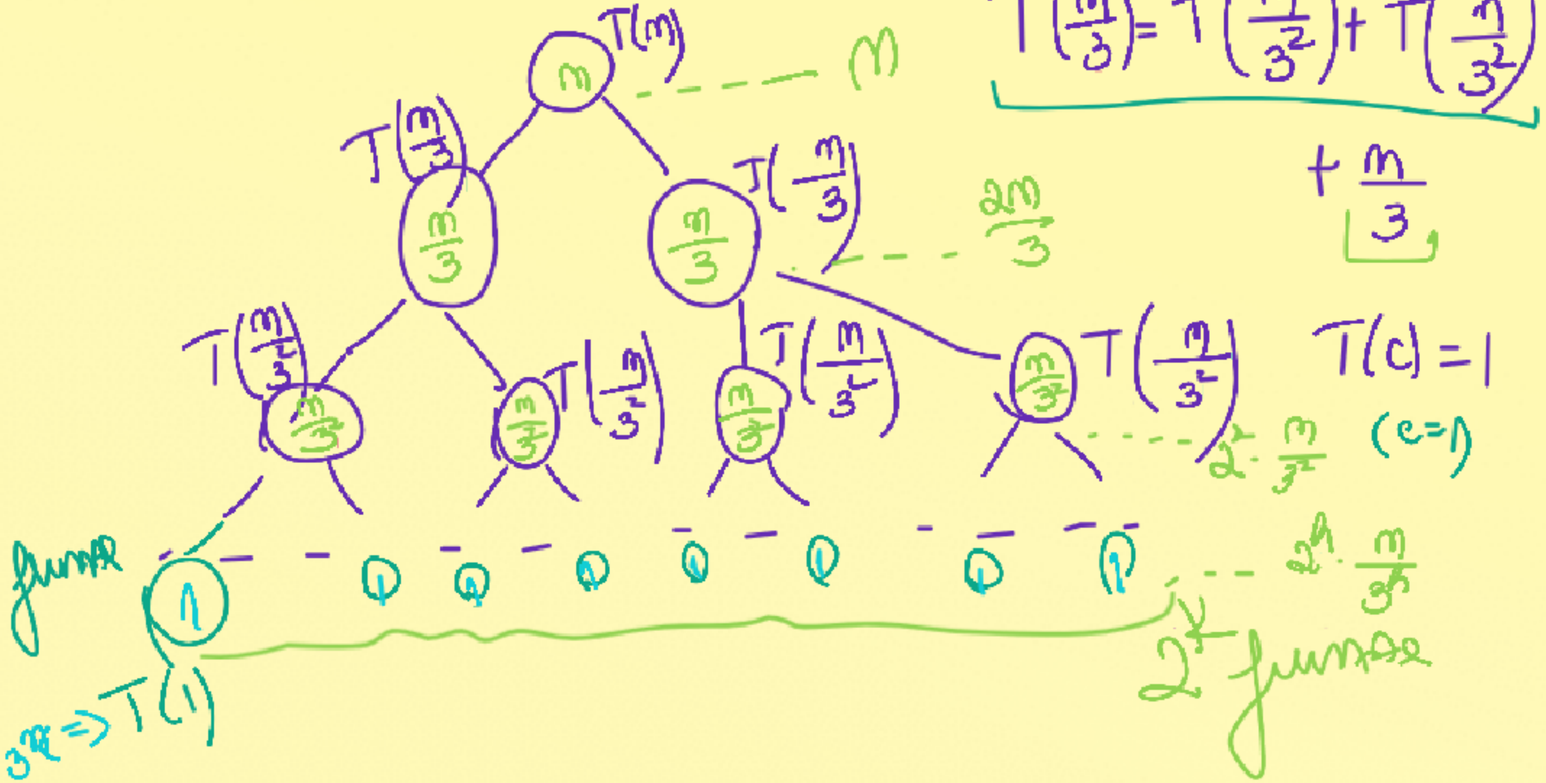
$1 + a + a^2 + \ldots + a^m = \frac{a^{m+1} - 1}{a - 1}$

$2^{k+1} = 2^{\log\left(\frac{\log n}{\log 2}\right)} = \frac{\log n}{\log 2} = \left(\frac{1}{\log 2}\right) \log n$

constant

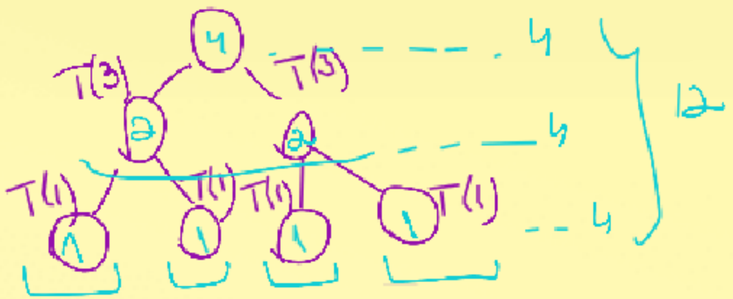
$2^{\log a} = a^{\log 2} = a$

$$T(n) = T\left(\frac{n}{3}\right) + T\left(\frac{n}{3}\right) + n$$



$$T\left(\frac{n}{3}\right) = T\left(\frac{n}{3^2}\right) + T\left(\frac{n}{3^2}\right) + \frac{n}{3}$$

T(3), n=9



$\frac{n}{3^k} = 1 \Rightarrow 3^k = n \Rightarrow k = \log_3 n$

ex: $T(n) = T(\frac{n}{2}) + T(\frac{n}{2}) + n$

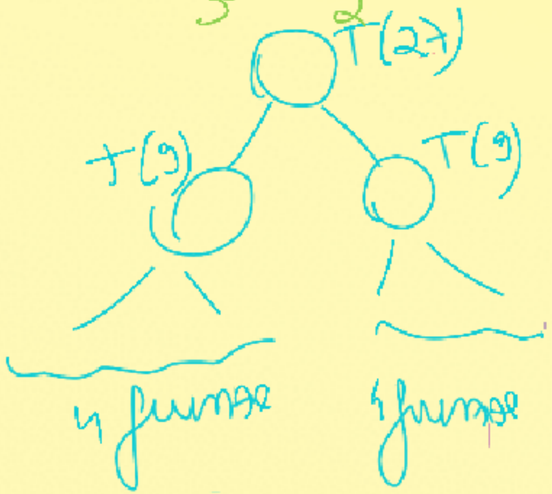
$\frac{n}{2^k} = 1 \Rightarrow k = \log n$

$n = 3 \Rightarrow$ 2 frunze



$n = 9 \Rightarrow$ 4 frunze

$n = 27 \Rightarrow$ 8 frunze



$$T(n) = \begin{cases} \\ \\ \end{cases} \sum_{k=0}^{h} 2^k \cdot \frac{n}{3^k}$$

$h = \log_3 n$

$$T(n) = \sum_{k=0}^{\log_3 n} \left(\frac{2}{3}\right)^k n$$

$$= n \cdot \sum_{k=0}^{\log_3 n} \left(\frac{2}{3}\right)^k$$

$$\frac{\left(\frac{3}{3}\right)^{\log_3 n + 1} - 1}{\frac{2}{3} - 1}$$

$$\sum_{i=0}^{n} a^i = \frac{a^{n+1} - 1}{a - 1}$$